



**NANYANG
TECHNOLOGICAL
UNIVERSITY**
SINGAPORE

Lecture 2: CA6121 AI in Project Management

Project Integration, Scope Development, PMP, Contracts, AI tools in PM

Presented by Dr Ravish Maqsood





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OBJECTIVES

To learn and understand:

- Project Integration.
- Project Life Cycle Models.
- Project Management Plan Development.
- Change Management.
- AI tools in Project Management.

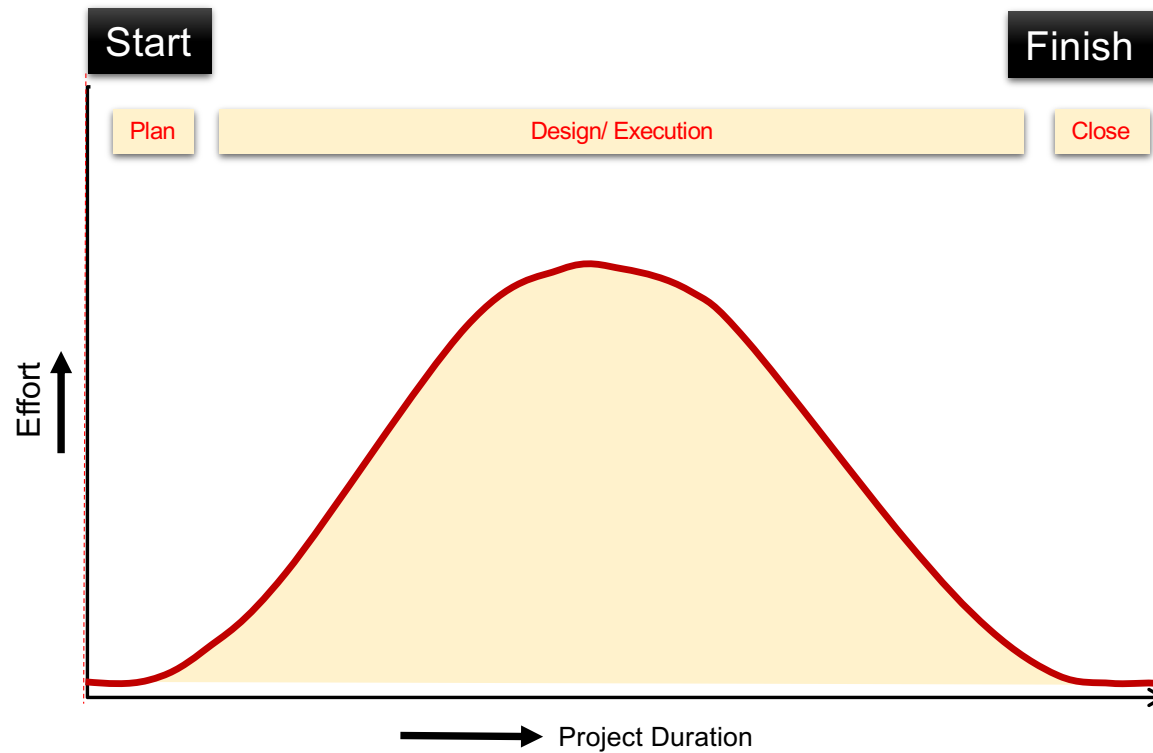


Project Integration/ Life Cycle

Project Management

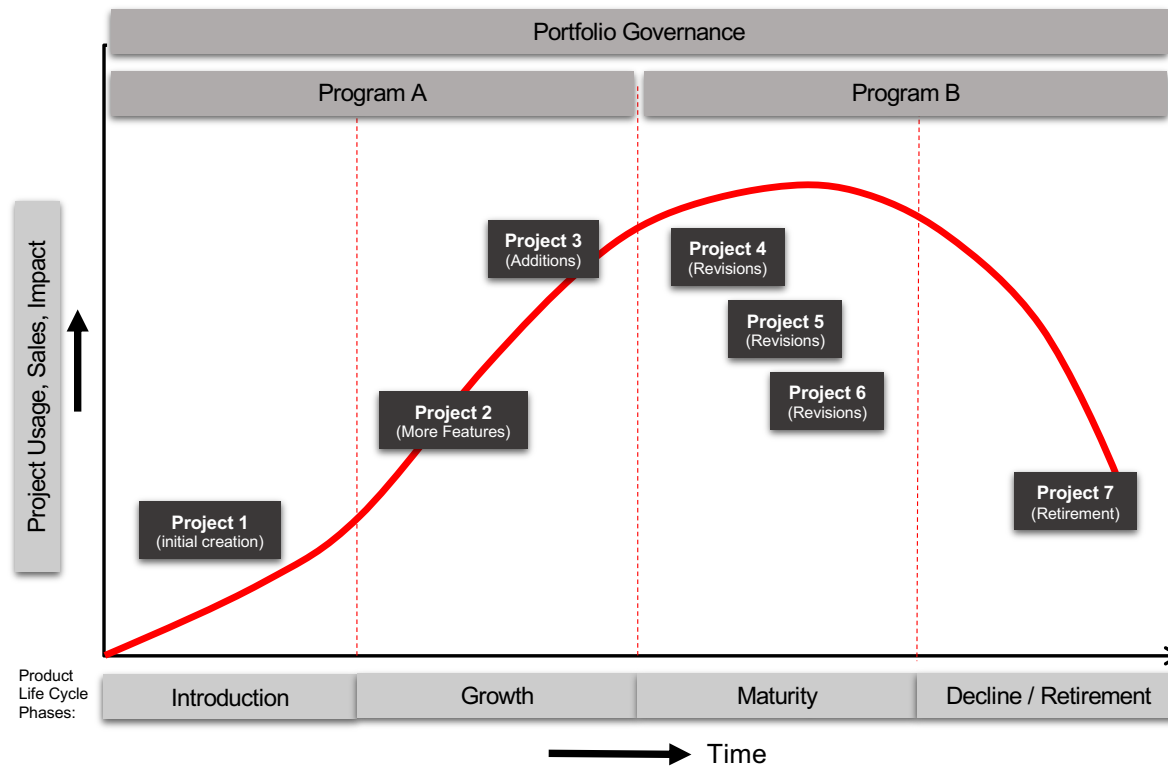


Project Life Cycle – Simplified Approach





Project Life Cycle – Simplified Approach

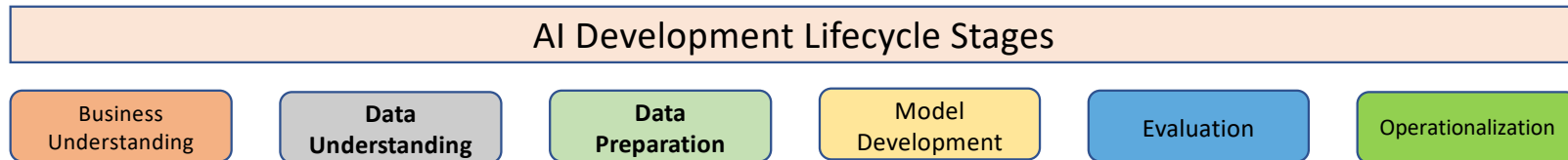


Source Reference: PMBOK 7th edition Figure 2-4. Sample Product Life Cycle



AI Development Projects

Recap



Project Manager could be from either of the disciplines:

- Computer Science
- Data Science
- Information Technology
- Other discipline with knowledge and interest in data



Project Management: Six Principles (PMBOK 8th Addition)

Collectively, these principles describe the mindset of project management, which in turn guides the mechanics of project management.

1. Adopt a Holistic View.
2. Focus on Value.
3. Embed Quality Into Processes and Deliverables.
4. Be an Accountable Leader.
5. Integrate Sustainability Within All Project Areas.
6. Build an Empowered Culture.



Project Management: Six Principles (PMBOK 8th Addition)

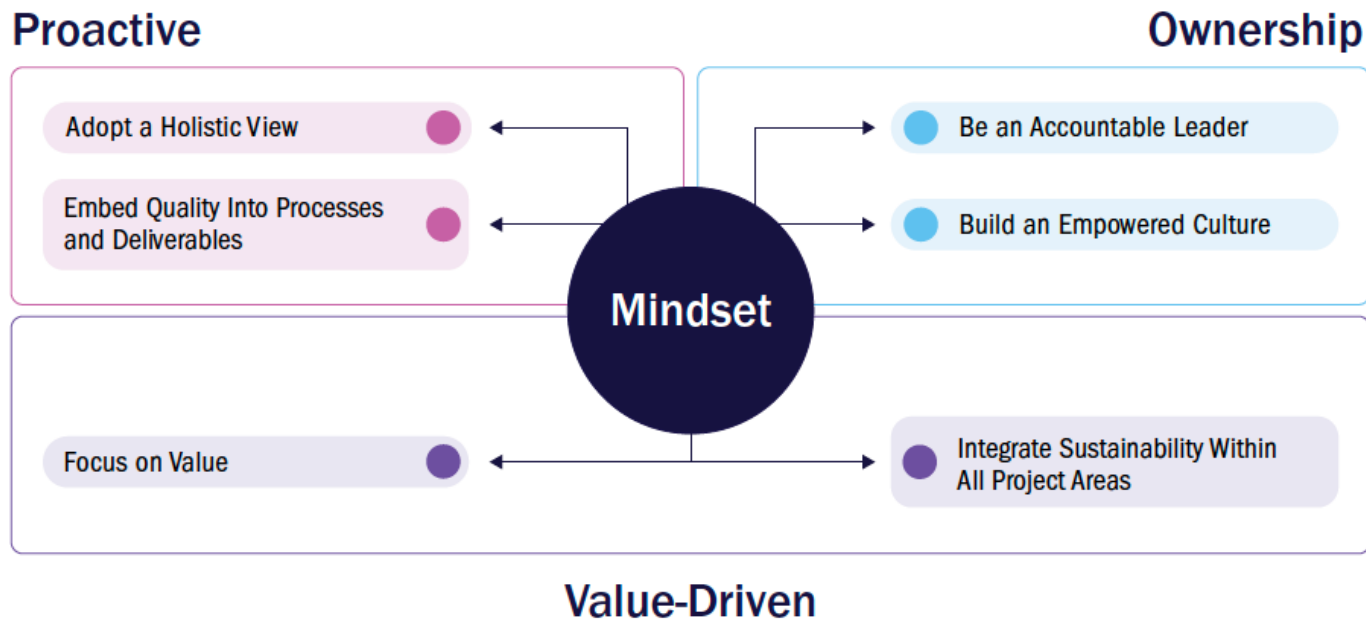


Figure 3-1. The Project Management Mindset



Project Life Cycles: Five Factors (PMBOK 8th Addition)

1. Project phases.
2. Project development approaches (predictive, adaptive, hybrid).
3. Considerations for selecting a development approach.
4. Delivery cadence.
5. Project Management Focus Areas (formerly known as Process Groups).



Project Life Cycles: Five Factors (PMBOK 8th Addition)

1. Project phases

- The phases may be **sequential, transitional, or overlapping**.
- In some projects a **combination of these phases** may be used, and each phase may include one or more stages of the project life cycle



Project Life Cycles: Five Factors (PMBOK 8th Addition)

2. Project development approaches

- A. Predictive.
- B. Adaptive.
- C. Hybrid.

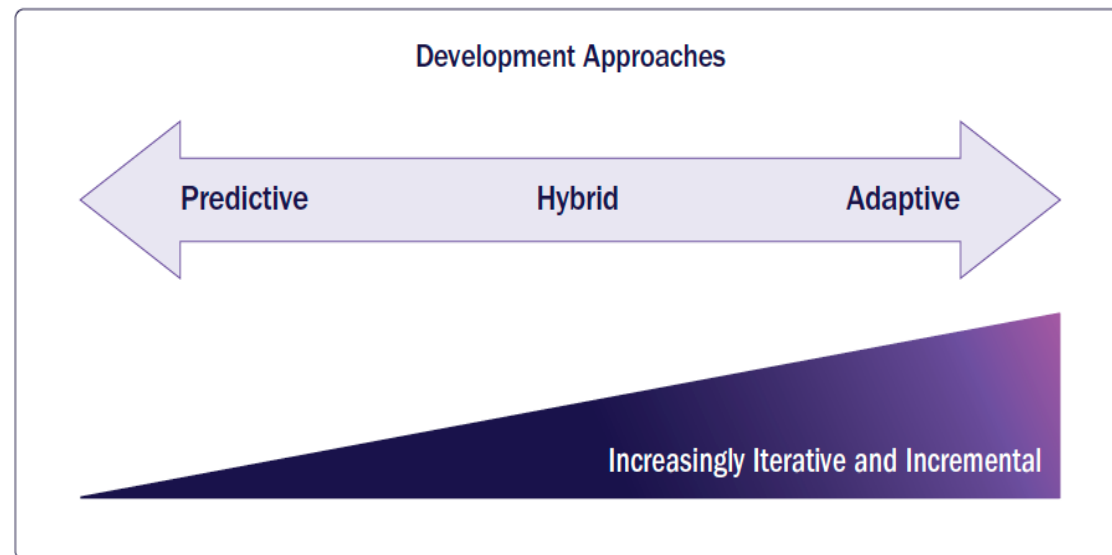


Figure 4-2. The Spectrum of Development Approaches



Project Life Cycles: Five Factors (PMBOK 8th Addition)

2. Project development approaches

A. Predictive Approach

Adopted when requirements are clear and available early in the project.

B. Adaptive Approach

Adopted when requirements are less clear early in the project, or the project is expected to incur more complexity and uncertainty.



Project Life Cycles: Five Factors (PMBOK 8th Addition)

2. Project development approaches

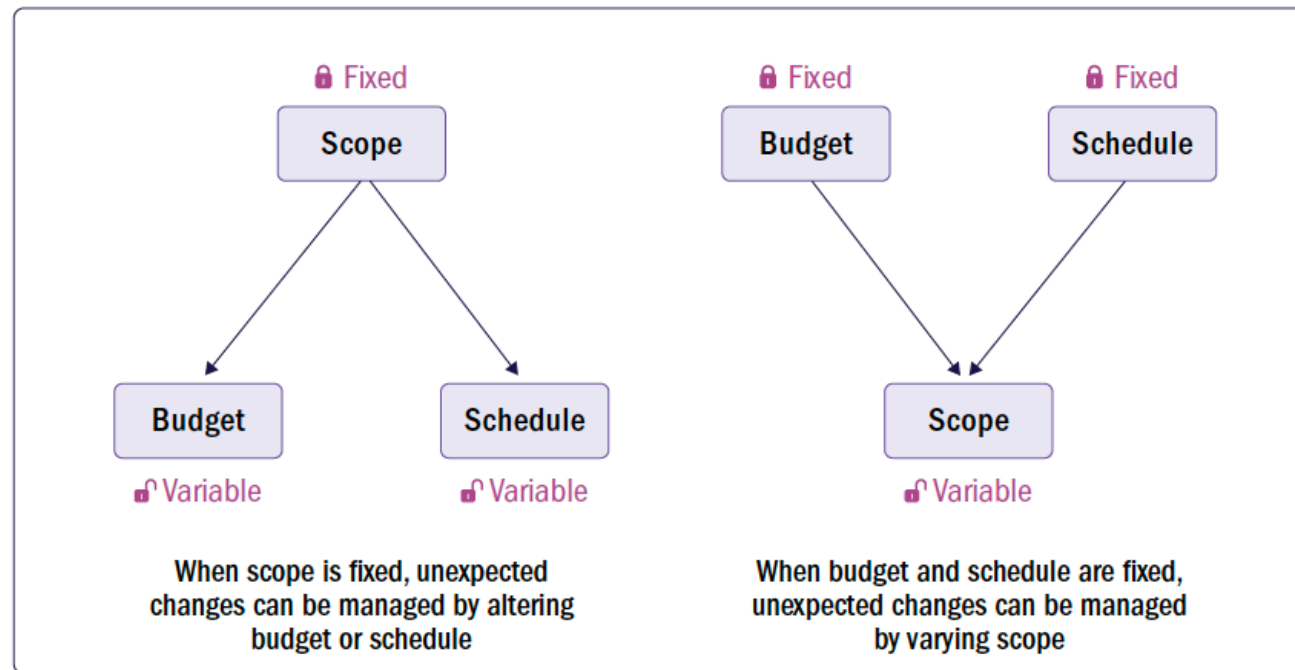


Figure 4-3. Constraint Options for Adaptive Approaches



Project Life Cycles: Five Factors (PMBOK 8th Addition)

2. Project development approaches

A. Predictive Approach

- Adopted when requirements are clear and available early in the project.
- The scope, schedule, cost, resource needs, quality requirements, and risks can be well defined in the early phases of the project life cycle and are expected to remain relatively stable.
- Waterfall / Traditional approach.



Project Life Cycles: Five Factors (PMBOK 8th Addition)

2. Project development approaches

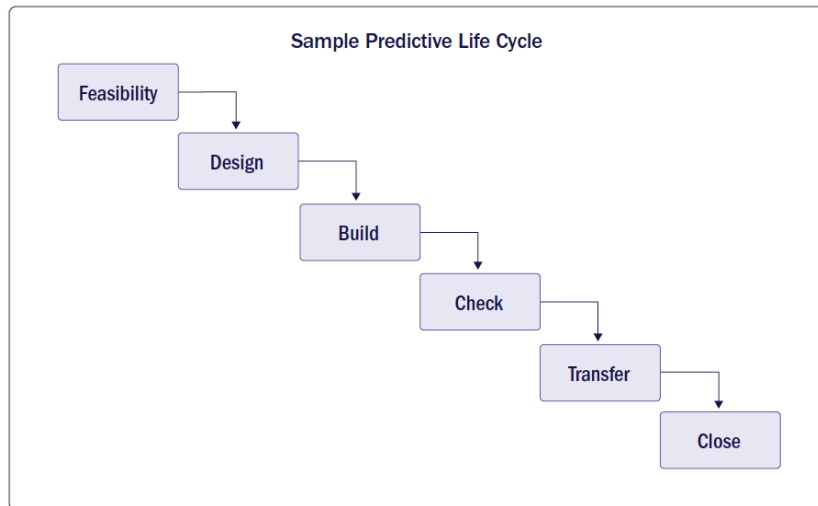


Figure 4-4. Sample Predictive Life Cycle

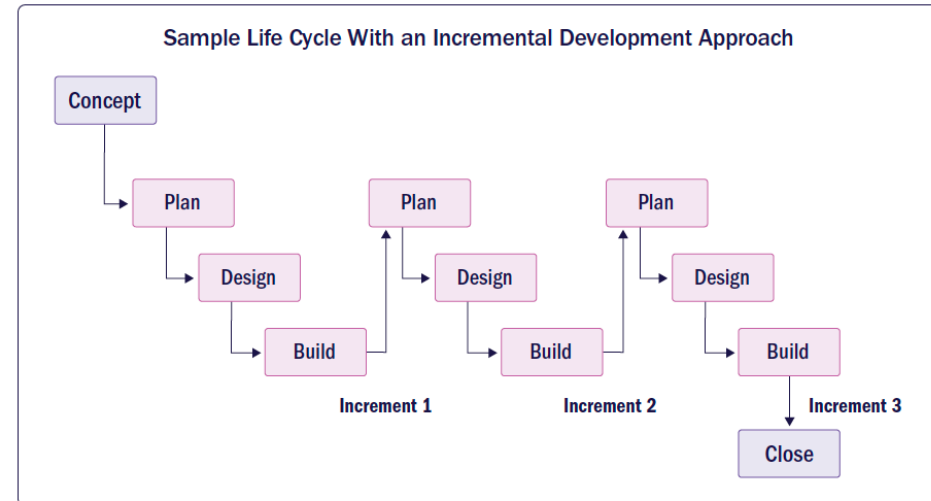


Figure 4-5. Sample Predictive Life Cycle With an Incremental Delivery



Project Life Cycles: Five Factors (PMBOK 8th Addition)

2. Project development approaches

B. Adaptive Approach

- Adopted when requirements are less clear early in the project, or the project is expected to incur more complexity and uncertainty.
- High-level of uncertainty and volatility and are likely to change significantly throughout the project.
- Agile Approach.



Project Life Cycles: Five Factors (PMBOK 8th Addition)

2. Project development approaches

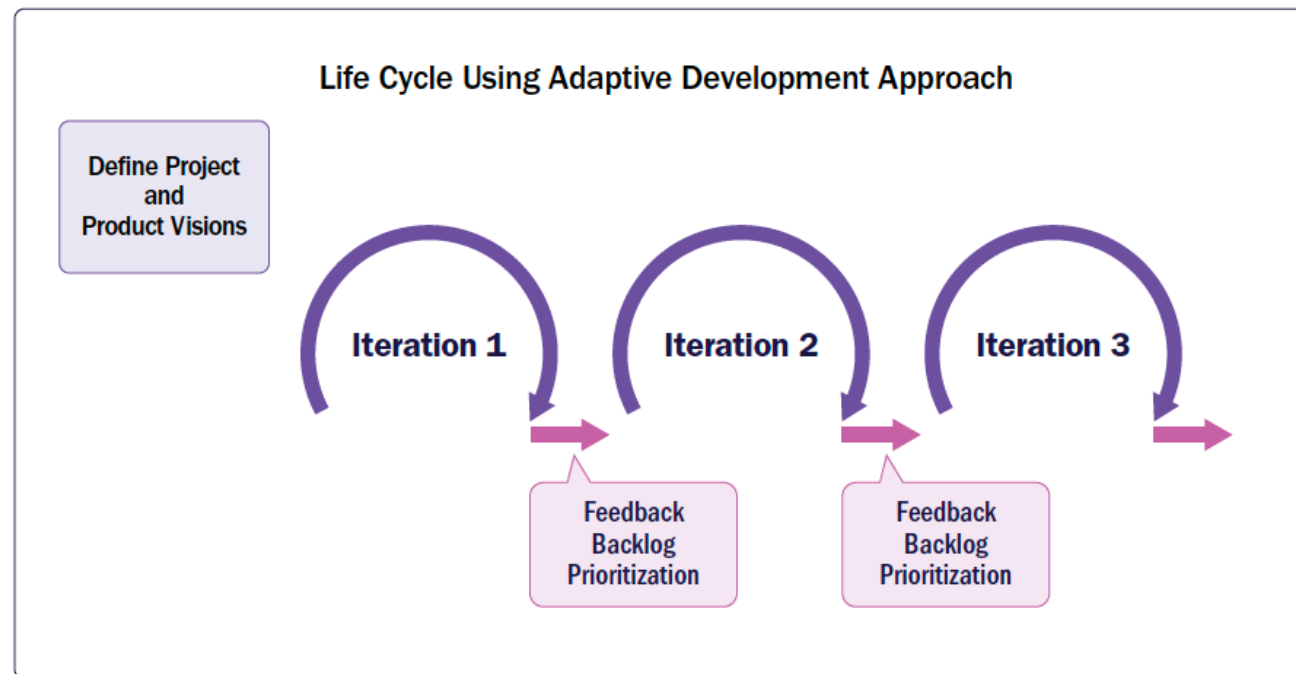


Figure 4-6. Life Cycle With an Adaptive Development Approach



Project Life Cycles: Five Factors (PMBOK 8th Addition)

2. Project development approaches

C. Hybrid Approach

- Mix of Predictive and Adaptive approach.
- Applied as combination of above for different phases of project.
- Mix of Waterfall and Agile Approach.



Project Life Cycles: Five Factors (PMBOK 8th Addition)

3. Considerations for selecting a development approach

- Several factors influence the selection of a development approach.
- Divided into categories of the:
 - Deliverables.
 - Project.
 - Organization.



Project Life Cycles: Five Factors (PMBOK 8th Addition)

4. Delivery cadence

- Based on the selected development approach, projects can have:
 - a single delivery.
 - multiple deliveries.
 - periodic deliveries.
 - continuous deliveries.



Structure of PMBOK 8th Addition

5. Project Management Focus Areas

- Project Management Focus Areas:
 - Initiating
 - Planning
 - Executing
 - Monitoring and Controlling
 - Closing



Project Management Performance Domains (PMBOK 8th Addition)

Seven (7) Project Management Performance Domains:

1. Governance
2. Scope
3. Schedule
4. Finance
5. Stakeholders
6. Resources
7. Risk



Project Management Performance Domains (PMBOK 8th Addition)

Table 2-1. Mapping Project Management Focus Areas to Performance Domains

Performance Domains	Project Management Focus Areas				
	Initiating Focus Area	Planning Focus Area	Executing Focus Area	Monitoring and Controlling Focus Area	Closing Focus Area
Governance	Initiate Project or Phase	- Integrate and Align Project Plans - Plan Sourcing Strategy	- Manage Project Execution - Manage Quality Assurance - Manage Project Knowledge	- Monitor and Control Project Performance - Assess and Implement Changes	Close Project or Phase
Scope		- Plan Scope Management - Elicit and Analyze Requirements - Define Scope - Develop Scope Structure		- Monitor and Control Scope - Validate Scope	
Schedule		- Plan Schedule Management - Develop Schedule		Monitor and Control Schedule	
Finance		- Plan Financial Management - Estimate Costs - Develop Budget		Monitor and Control Finances	

Table 2-1. Mapping Project Management Focus Areas to Performance Domains

Performance Domains	Project Management Focus Areas				
	Initiating Focus Area	Planning Focus Area	Executing Focus Area	Monitoring and Controlling Focus Area	Closing Focus Area
Stakeholders	Identify Stakeholders	- Plan Stakeholder Engagement - Plan Communications Management	- Manage Stakeholder Engagement - Manage Communications	- Monitor Stakeholder Engagement - Monitor Communications	
Resources		- Plan Resource Management - Estimate Resources	- Acquire Resources - Lead the Team	Monitor and Control Resourcing	
Risk		- Plan Risk Management - Identify Risks - Perform Risk Analysis - Plan Risk Responses	Implement Risk Responses	Monitor Risks	



Project Integration Management

Project integration is one of the most important aspect in Project Management

- Identify the type of project
- Understand the needs on the project
- Select the right approach
- Understand interdependencies, constraints and assumptions
- Gateways

Predictive and Planned

Hybrid

Incremental and Iterative

Traditional Approach

Combination

Agile Approach



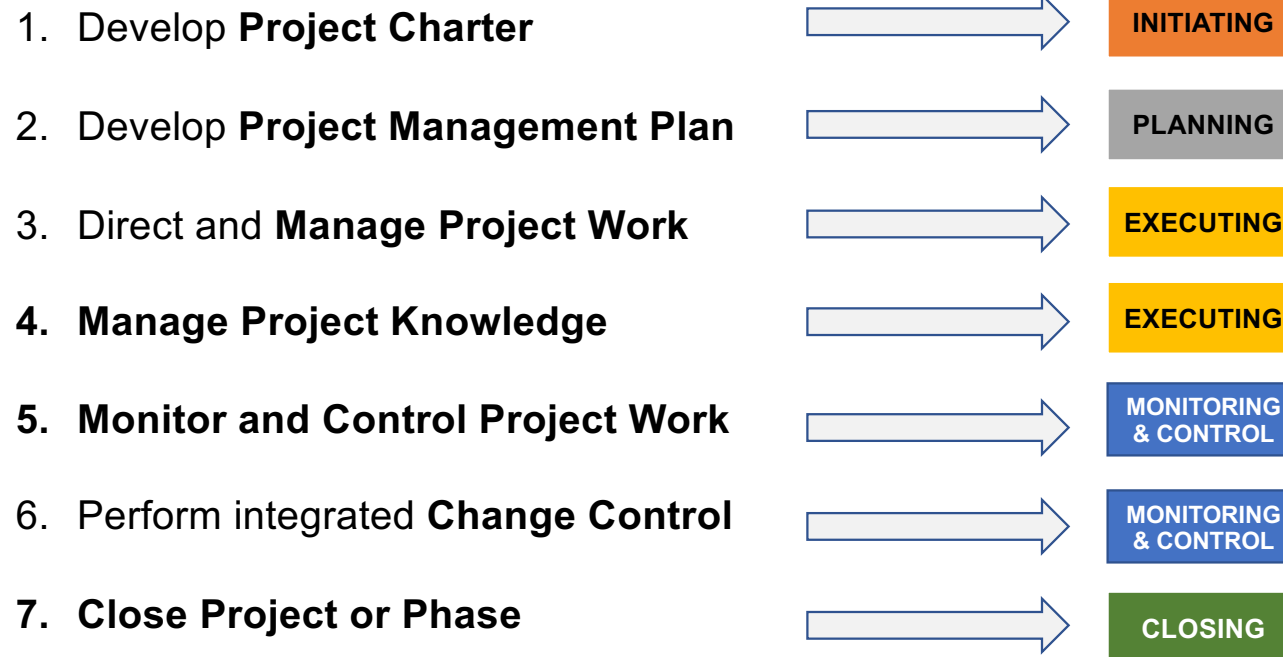
Project Integration Management

- Project Manager takes the lead.
- Client, Consultants and SME* are involved.
- Industry updates and team's experience.
- **Project Manager must read the project documents** available prior starting to set the Project Charter, and Project Management Plan.
- Do not assume, always check the contractual agreement.

SME* - Subject Matter Expert



Project Integration Management - Processes





Project Integration Management

PMBOK® Guide approach:

1. Inputs

- Enterprise Environment Factors
- Organizational Process Assets

2. Tools & Techniques

- Meetings
- Expert Judgements
- Decomposition
- Others

3. Outputs

- Drawings, Specifications,
- Documentation
- Reports
- Others





Develop Project Charter

a. Project Charter consist of business/ customer needs, assumptions, constraints, high-level requirements of new product/ service or result.

- Project Purpose
- Objectives
- Requirements
- Assumptions & Constraints
- Description and boundaries
- High level risks
- Key Milestones
- Budget Summary
- Stakeholder list

b. Assumption Log



Develop Project Charter

SAMPLE

Introduction – Mention a brief about the project.

Project objectives and success criteria – mention project objectives or goals as set by the contract and any specific requirements.

Project scope, deliverables and requirements – describe project scope summary, boundaries, and requirements. A detailed deliverable list to be provided as per the contractual agreement.

Budget – the cost estimate for the project.

Schedule/milestones – project milestones, or stages, for measuring its progress and success.

Constraints and assumptions – detail the known and unknown constraints and assumptions.

Summary of risks – list all the known possible risk to the project.

Team and organization – mention key people and stakeholders with their roles and contact details. Attach an organization chart to show their reporting.

Approvals – ensure written endorsed approval from the stakeholder authorized personnel on the project charter document.



Develop Project Management Plan



PLANNING

a. Project Management Plan

- Subsidiary Management Plans:
 - Scope Management Plan
 - Project Communication Plan
 - Project Engineering Management
 - Project Quality Plan
 - Project Control Plan (Resource, Cost & Schedule)
 - Project Risk Management Plan
 - Project Procurement Plan
 - Project Safety Plan
 - Stakeholder Management Plan
- Baselines:
 - Scope baseline
 - Schedule baseline
 - Cost baseline
- Additional Components:
 - Change Management plan
 - Reviews



Develop Project Management Plan



PLANNING

Table 4-1. Project Management Plan and Project Documents

Project Management Plan	Project Documents	
1. Scope management plan	1. Activity attributes	19. Quality control measurements
2. Requirements management plan	2. Activity list	20. Quality metrics
3. Schedule management plan	3. Assumption log	21. Quality report
4. Cost management plan	4. Basis of estimates	22. Requirements documentation
5. Quality management plan	5. Change log	23. Requirements traceability matrix
6. Resource management plan	6. Cost estimates	24. Resource breakdown structure
7. Communications management plan	7. Cost forecasts	25. Resource calendars
8. Risk management plan	8. Duration estimates	26. Resource requirements
9. Procurement management plan	9. Issue log	27. Risk register
10. Stakeholder engagement plan	10. Lessons learned register	28. Risk report
11. Change management plan	11. Milestone list	29. Schedule data
12. Configuration management plan	12. Physical resource assignments	30. Schedule forecasts
13. Scope baseline	13. Project calendars	31. Stakeholder register
14. Schedule baseline	14. Project communications	32. Team charter
15. Cost baseline	15. Project schedule	33. Test and evaluation documents
16. Performance measurement baseline	16. Project schedule network diagram	
17. Project life cycle description	17. Project scope statement	
18. Development approach	18. Project team assignments	

Caution: Not every item is applicable for all projects

Reference from PMBOK 6th Edition



Develop Project Management Plan



PLANNING

Project Baseline (Scope, Schedule, Cost)

- Project Manager works with the team to **firm up the scope baseline**.
- Sometimes the “**as sold**” scope may differ to what project documents received.
- Project control lead works on the **baseline schedule, and cost** for the project.
- This shall be agreed with the client and to be used from **here onwards to monitor and control** the project.

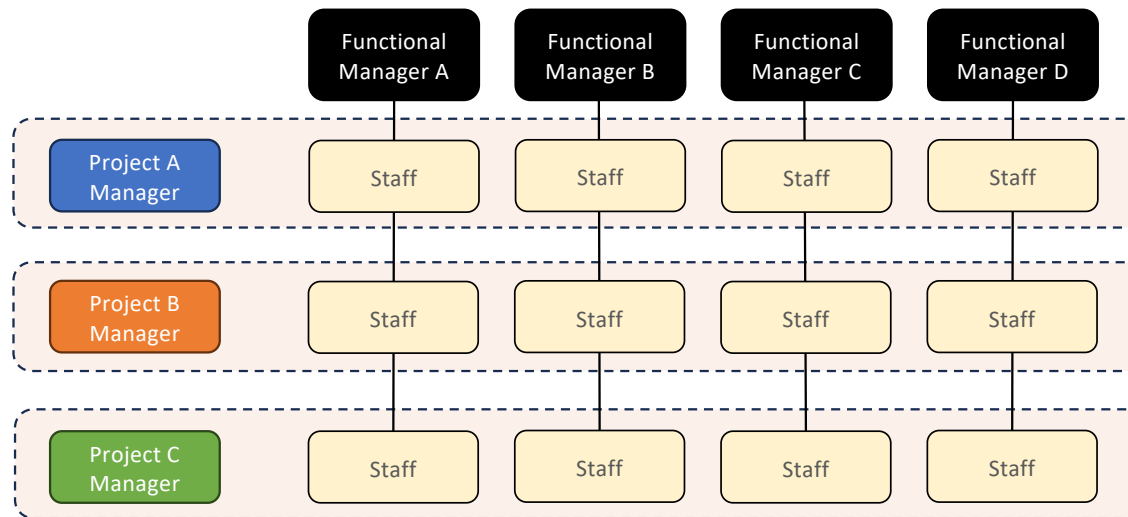


Develop Project Management Plan



PLANNING

Project Organization Structure



Matrix Organization



Develop Project Management Plan



PLANNING

Project Communication Plan

- This is a key communication protocol document used throughout the life of the project.
- **Organization chart** is prepared with names and their contact details.
- **Approval matrix** for all Technical & Commercial deliverables.
- **Authorization matrix** to ensure the visibility on who can authorize payments, changes, etc.



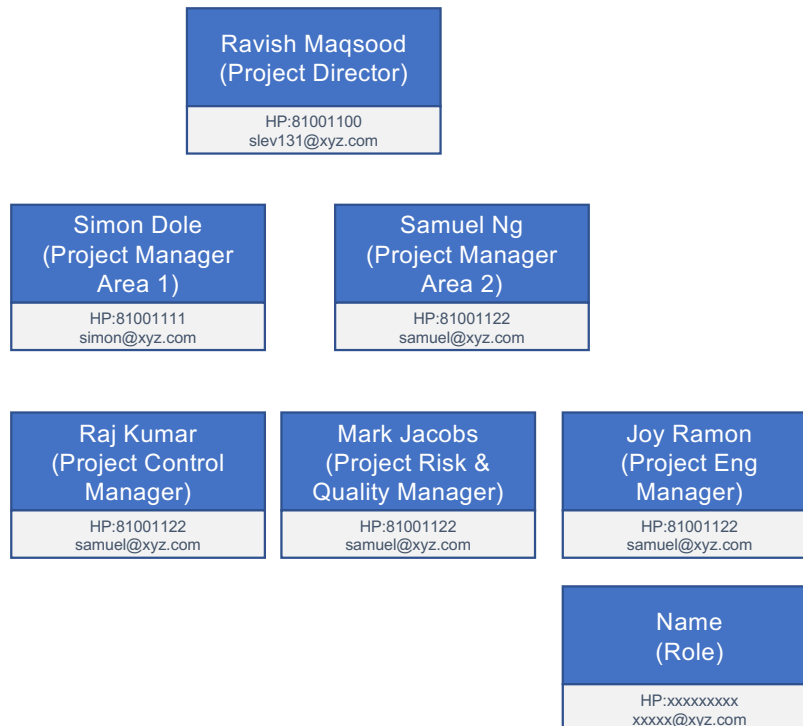
Develop Project Management Plan



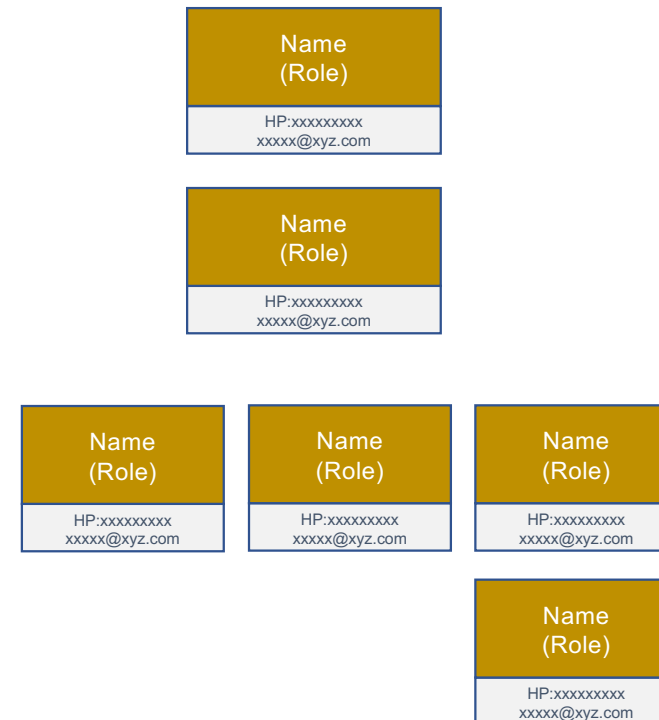
PLANNING

Project Communication Plan

Client/ Customer Organization



Contractor Organization





Develop Project Management Plan



PLANNING

Project Engineering Management

- **Engineering coordination details** mentioned.
- **Approval system.**
- **Templates** to be used for deliverables.
- **Numbering system.**
- **Competency matrix** of the team.
- **Approval and authority** matrix.



Develop Project Management Plan



PLANNING

Project Quality Plan

- Overall Quality management details mentioned.
- Approval system for any project procedures/ templates to be used on the project.
- Templates to be used for quality deliverables.
- Quality testing, approvals, trainings.
- Quality Assurance and Quality Control Plan.
- Audit Schedule, and other related templates.



Develop Project Management Plan



PLANNING

Project Control Plan (Resource, Cost, & Schedule)

- Overall Project Control Management details.
- Project reporting schedule, templates.
- Project scope, cost, schedule monitoring for all activities.



Develop Project Management Plan



PLANNING

Project Risk Management Plan

- Overall Project Risk Management details.
- Project risk meetings schedule, reports, templates.
- Risk Register, Risk Matrix etc. to be mentioned.



Develop Project Management Plan



PLANNING

Project Procurement Plan

- Overall Project Procurement Management details.
- Required purchases; cloud, servers, licenses, etc.
- Project Vendors Contracts, and Sub-Contracts management.
- Project vendor inspection, expediting, acceptance test.
- Installation progress and payment management.



Develop Project Management Plan (PMP)



PLANNING

(Sample)

Template for Project Management Plan

Project Name: *(Insert Project Name)*

Project No:

Version History: *(Version No, Date, Author, Key Differences)*

Content

- Introduction
- Management Approach
- Roles and Responsibilities
- Integration Management Processes
 - Create Project Charter
 - Develop the Scope Statement
 - Develop Project Management Plan
 - Direct and Manage Project Work
 - Monitor and Control Project Work
 - Perform Integrated Change Control
 - Finalize and Close Project or Phase
- Assumptions
- Issues and Risks
- Plan Approval

Attachments

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Direct & Manage Project Work



EXECUTING

- a. Deliverables
- b. Work Performance data
- c. Issue log
- d. change requests
- e. Project Management Plan updates
- f. Organizational Process assets updates



Manage Project Knowledge



EXECUTING

a. Lessons register

b. Project Management Plan updates

- Work Performance Reports: -Project Fortnightly/Monthly Progress Report (All disciplines)
- Updates and Report.-Justification-Supporting docs-Updates
- Update all other associated Plans

c. Organizational Process assets updates



Monitor and Control Project Work



MONITORING
& CONTROL

- a. Work Performance Reports
 - Fortnightly and Monthly Progress Reports.
- b. Change Requests
 - Supporting documents to be maintained.
 - Change log with details.
- c. Project Management Plan Updates
- d. Project Documents updates
 - Cost forecasts
 - Issue Log
 - Lessons register
 - Risk Register



Perform Integrated Change Control



MONITORING
& CONTROL

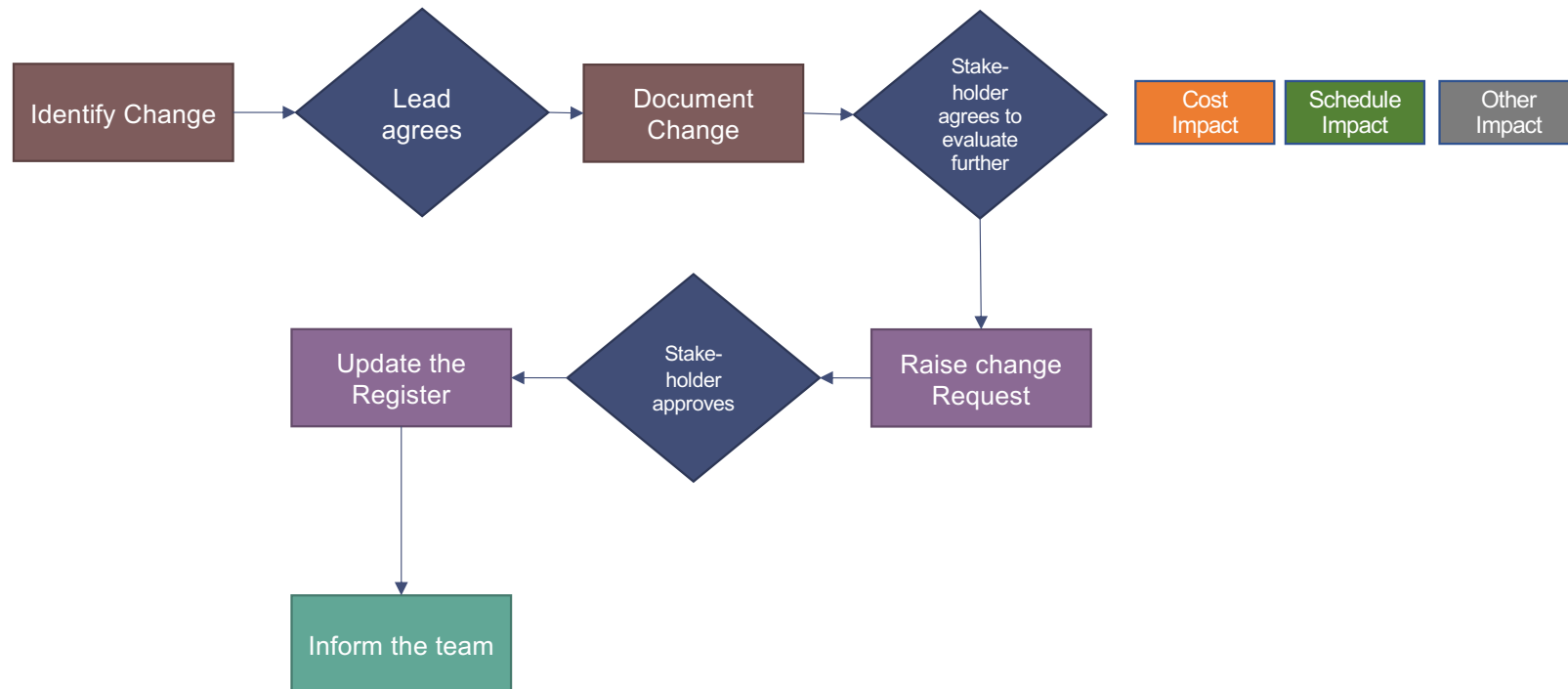
- a. Approved Change Requests
- b. Project Management Plan updates
- c. Project Documents updates



Perform Integrated Change Control



MONITORING
& CONTROL





Close Project or Phase



CLOSING

- a. Project Documents updates
- b. Final Product
- c. Final Report
- d. Organizational Process assets updates



Close Project or Phase



CLOSING

Project Management Plan and associated plans (Work Performance):

- Accepted deliverables
 - Agreed documents
 - Technical & Commercial Data
 - Records
-
- Consolidate all the required information required for the close out as per the contractual agreement.
 - Close all vendor, sub-contract, third party contracts.
 - Demobilize the team as per the Resource Plan.



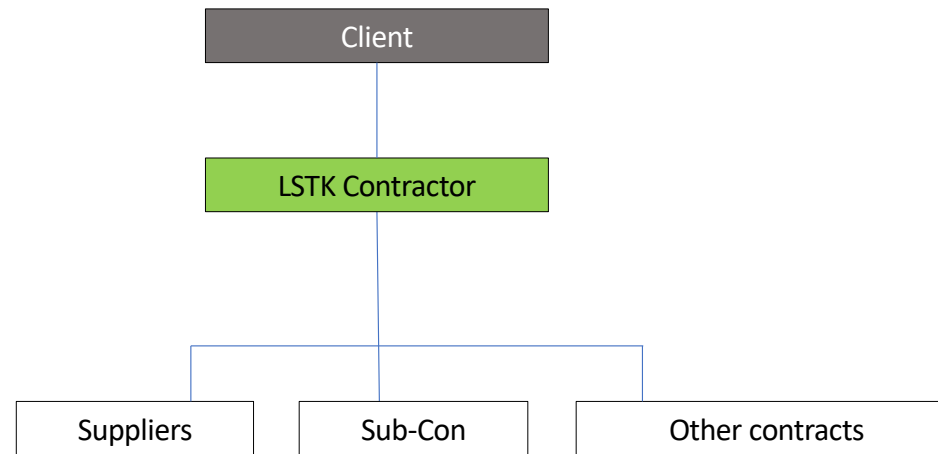
Project Execution Contracts

High level Perspective



Project Execution Strategies: LSTK Approach

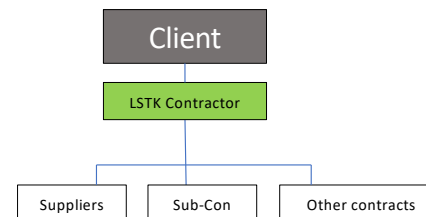
LSTK: lump sum turnkey





Project Execution Strategies: LSTK Approach

- Contractor taking project on LSTK basis is responsible to execute:
 - Engineering (Design, authority clearance etc.)
 - Procurement (Purchasing of equipment, material and sub-contracts)
 - Self performs construction
- Commissioning / Deployment / Evaluation / Validation
- Scope is to turn the project deliverables operational meet the specific output as per the design criteria

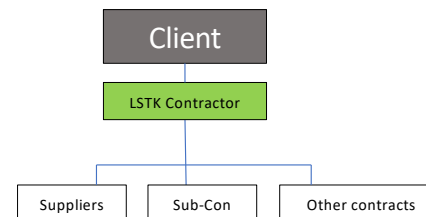




Project Execution Strategies: LSTK Approach

Other key pointers:

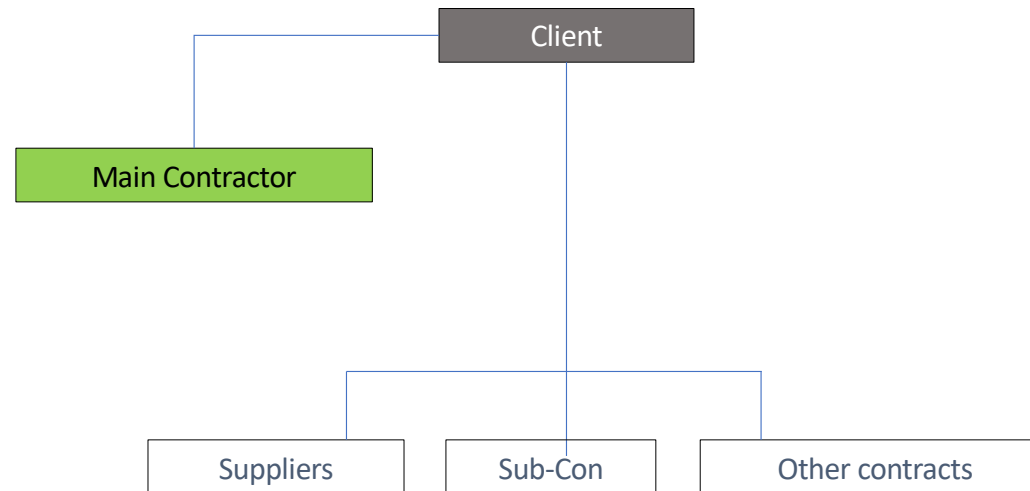
- Contractor holds responsibility of budget and schedule
- Contractor holds contract with sub-contractors
- Liable for any defective equipment or performance
- Contractor has high risk
- Contractor makes high profit margins in respect to TIC of the project.





Project Execution Strategies: Reimbursable (RR) Approach

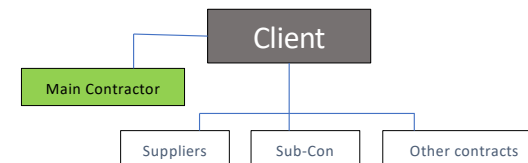
Engineering, Procurement and Project Management





Project Execution Strategies: Reimbursable (RR) Approach

- Contractor taking project on RR basis is responsible to execute:
 - Engineering (Design, authority clearance etc.)
 - Procurement (Purchasing of equipment, material and sub-contracts)
 - Act on behalf of the client for development and deployment

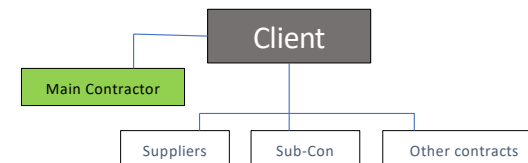




Project Execution Strategies: Reimbursable (RR) Approach

Other key pointers:

- All risks with the client
- Budget and schedule slippage
- Client holds contract with sub-contractors
- Contractor renders their services to execute the project
- Contractor has less risk
- Contractor has less profit as service fee is very less in respect to TIC of the project.





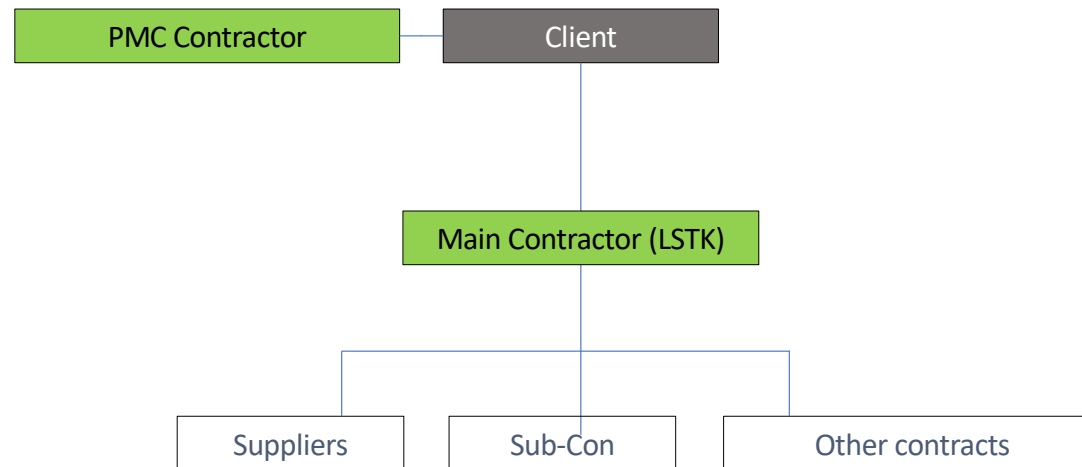
Project Execution Strategies: LSTK vs Reimbursable (RR)

	LSTK basis Contract	Reimbursable (RR) basis Contract
Accountability	Contractor fully accountable	Client has multiple points of accountability
Risk	Contractor holds risk	Client holds risk
Schedule	Fixed date of completion	No fixed completion schedule
Price	Fixed price contract	Schedule of Rates Cost Plus
Procurement	Contractor responsibility	Manage the process and act as an agent for the client
Quality/ Performance Guarantee	Contractor guarantees performance of completed facility	Contractor does not provide performance guarantees
Client's involvement	Contractor in control	Client in control
Defective works / Services	Contractor to rectify any defects	Assists client to manage the rectification of defects
Profit margins for Contractor	Higher profits	Only charges the service fee and potential savings goes to client.



Project Execution Strategies: Project Management Consultancy (PMC) Approach

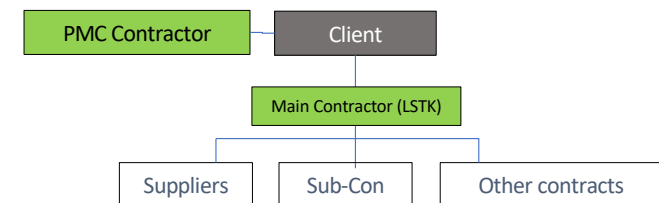
PMC: Project Management Consultancy





Project Execution Strategies: Project Management Consultancy (PMC) Approach

- Large sized projects need to have stronger client's team to oversee the project
- Client engages PMC contractor to support to project execution on their behalf
- PMC is responsible to:
 - set the basis of design,
 - specifications,
 - technical documentation,
 - commercial documentation,
 - bidders list preparation,
 - tendering,
 - evaluation,
 - award and then management of the whole execution of the project.

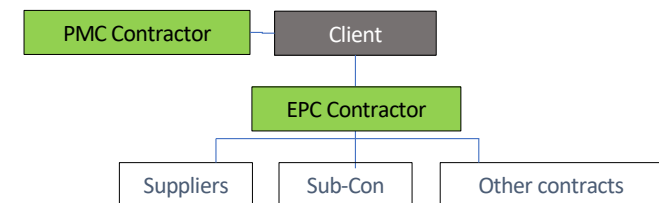




Project Execution Strategies: Project Management Consultancy (PMC) Approach

Other key pointers:

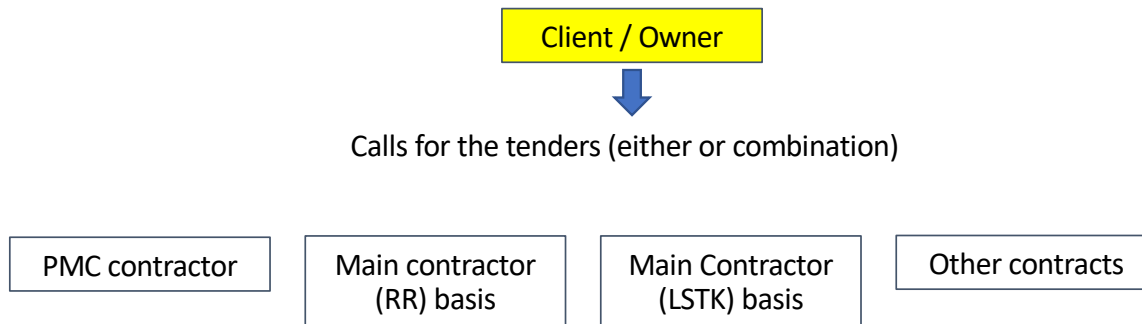
- Client do not need to hire more staff internally to execute the project.
- Takes advantage of the experience of the PMC contractor to execute the project





Project Execution Strategies: Tender

Once project is decided to take the next steps, client calls for the quotations/ tenders for various items:



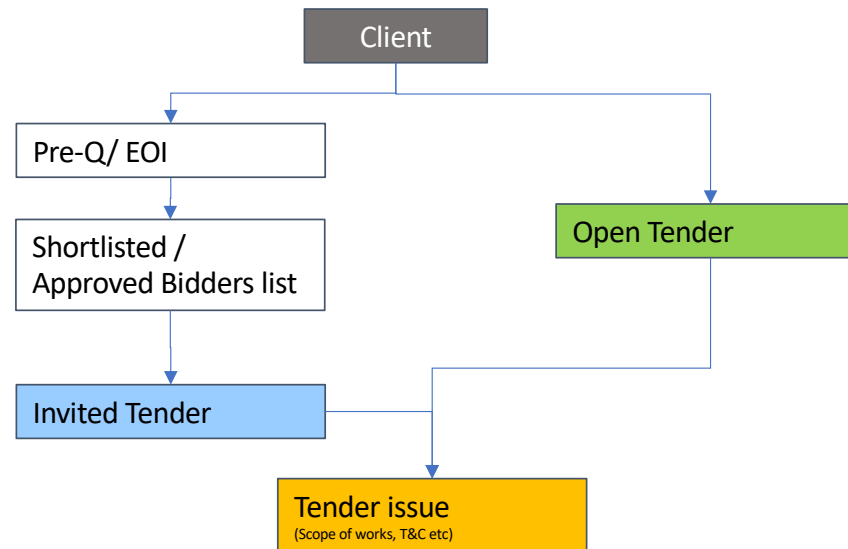


Project Execution Strategies: Tender

- Once the **bidders list** is shortlisted, then tender is called to invite the pre-selected tenderers.
- **Client calls for the tender** and invites **or** open tender to the public to submit their interest / Pre-Qualification / Bid
- Depending on the interest, the **contractor will prepare their bids**.
- **Contractors sets their execution strategies** to help them to prepare their best price.



Project Execution Strategies: Tender Process



EOI: Expression of interest
Pre-Q: Pre-Qualification



Project Execution Strategies: Tender Participation / Preparation

Key pointers:

- **Scope** of the project (Basic Design, EPC, Consultancy etc.)
- **Location** of the project (Country, state, etc.)
- **Availability of in-house capabilities & resources** to execute the proposed scope of work

- **Project timeline**
- **Project specific requirements**
- **Authority requirements**
- **Visa and immigration requirements**

- **Risks** and their mitigation
- Any **past-experience** with this client
- **Financial background checks**

Note*: Highlighted in red indicates where **AI can be useful**



Project Execution Strategies: Tender Participation / Preparation

Execution pointers:

- Past **lessons / experience** from similar client or project
- **Realistic schedule and budget**
- **Authority clearance and other permits**
- **Resources**
- Single location execution team or multi location
- Best **cost saving strategy** to win the job
- **Value** preposition (improvement to cost, schedule, operations)
- **Competitive advantage**
- **SWOT** analysis (Strength, Weakness, Opportunity, Threats)

Note*: Highlighted in red indicates where **AI can be useful**



Project Execution Strategies: Project Management Role

Project Execution

During Proposal Stage:

- Support the organization during bid submission
- Review the tender requirements and internal meetings
- Coordinate internally with the stake holders to prepare the best execution strategy
- Provide input to proposal with the technical write-up, required effort hours, execution strategy, proposed organization chart, key nominations with their CVs etc.

Note*: Highlighted in red indicates where AI can be useful



Project Execution Strategies: Project Management Role

Project Execution

During Execution Stage:

- Revisit proposed **execution strategy** (**re-align**, as necessary).
- **Kickoff meeting**.
- **Team selection** and set up.
- Conduct and participate in various **reviews**.
- Seek **necessary approvals** from the client and various stake holders.
- Prepare, provide inputs to detailed **project schedule**.
- **Execute the project**.

Note*: Highlighted in red indicates where **AI can be useful**



Project Execution Strategies: Project Management Processes

Project management processes:

1. Initiating

2. Planning

3. Executing

4. Monitoring and Controlling

5. Closing



Project Execution Strategies: Project Management Knowledge

Project management knowledge draws following areas:

1.Integration

6. Procurement

2.Scope

7. Human resources

3.Time

8. Communications

4.Cost

9. Risk management

5.Quality

10.Stakeholder management



Artificial Intelligence (AI)

High level Perspective



AI in Project Management: **Where can be implemented?**

- AI to be used as to **complement the human capabilities**.
- Not to focus on automating everything on a project using AI.
- Focus on routine repetitive tasks which maintaining active human oversight.



AI in Project Management: Human in the Loop Approach

- Identify key decision points where human input or validation is required.
- Clear roles and responsibilities for the team members.
- Regular reviews and audits.



AI in Project Management: Primary Barrier

- Management approval
- Technical knowledge
- Tool availability



Benefits

AI in Project Management



AI tools in Project Management: Benefits

Benefits of AI technologies for Project Managers

- Increased Productivity
- Improved Forecasting & Estimates
- Enhanced Decision Making
- Resource Management
- Innovation



AI tools in Project Management: Benefits

- Budget and finance.
- Task duration estimation.
- Risk mitigation.
- Resource allocation.
- Procurement and sub-contracts management.
- Lessons and claims management.
- Quality & safety management.
- Real-time based visibility and reporting.
- Improved communication
- Many more.



Challenges

AI in Project Management



AI tools in Project Management

Challenges of AI technologies for Project Managers

- Management approvals.
- Validation and verification through human intervention.
- Data availability.
- Quality of data.
- Ethics and governance.
- Bias.
- Integration & adaption.
- Skills required.
- Job security.



AI in Project Management: Challenges

- AI implementation requires data to be used which may be confidential.
- Reuse of the existing data to identify patterns.
- Limits creativity.
- Requires regular security audits.
- Implementation of proper authentication and access controls.
- Encryption requirements for any data storage or transmission.
- Set compliance requirements.



AI in Project Management: Challenges

- Human experience and knowledge is required to set-up a project.
- PM knowledge areas can not be fully replaced by AI.
- AI can automate or replace simple and repetitive tasks pattern.
- Procurement and quality management tasks are easy to automate.
- AI in project management still not fully matured.
- Traditional tools and excel sheets are commonly used.
- **Organization management is not receptive of the idea.**



AI in Project Management: Challenges (contd.)

- Lack of control on reporting & fear of losing control (sometimes PM has a reason to play with the numbers).
- Unawareness of the AI tools availability.
- Uncertainty about their effectiveness as not tested or proven in all sectors.
- Data privacy concerns.
- Data quality and availability.
- Transparency and trust about AI tools implemented.
- Algorithmic biased and ethical concerns.



AI in Project Management: Challenges (contd.)

- Too much noise about it in the market.
- Unclear how it will integrate to the existing systems.
- Lack of resources to implement, monitor, and train.
- May see as more costly and time consuming.
- No clarity on taking back control if AI fails.
- Many more reasons.



Artificial Intelligence (AI) in Project Management

Not clear on where to use AI tools



What it takes to use AI in Projects?

AI in Project Management



AI in Project Management

- Understanding of Project Management Workflow and where it adds value to use AI?
- AI can not be implemented on every workflow.
- Some of the areas can be used as:
 - Data consolidation, analysis, collation
 - Generating periodic communications
 - Auto response to regular queries
 - Planning
 - Reporting
 - Monitoring
 - Prediction
 - Others...



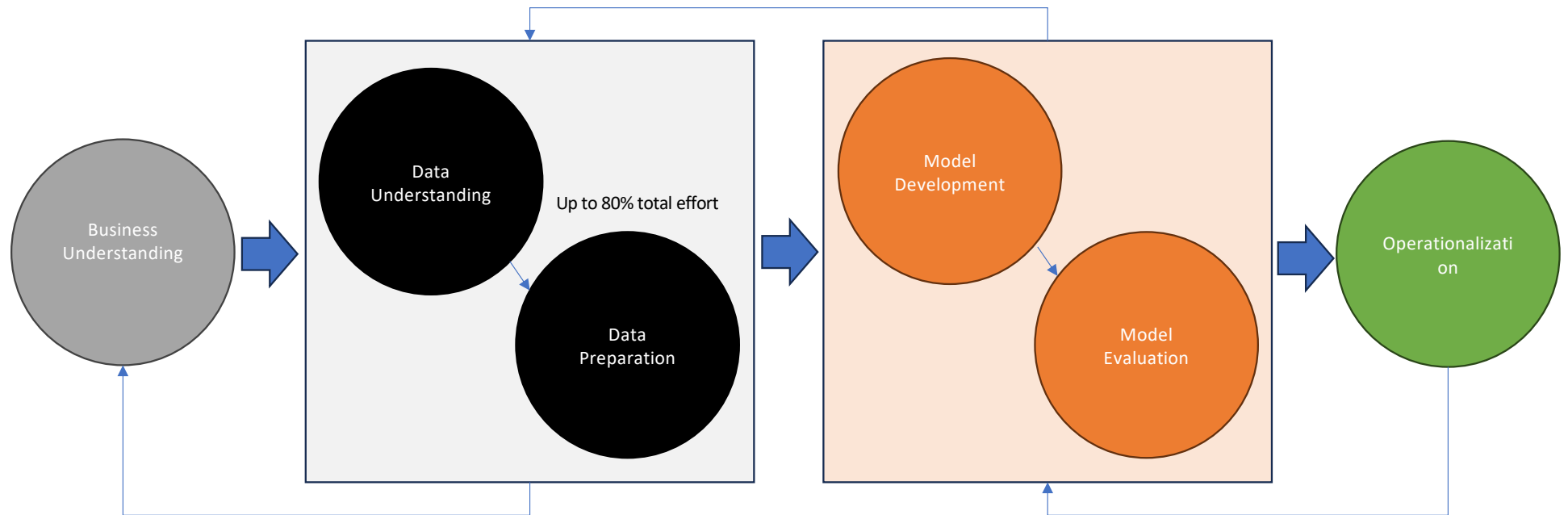
AI in Project Management: What is takes to implement AI?

Six (6) phases of implementation process of a AI project:

1. Business Understanding
2. Data Understanding
3. Data Preparation
4. Model Development
5. Evaluation
6. Operationalization



AI tools in Project Management: General Workflow





AI in Project Management: Pitfalls

Data quality and availability

Finding relevant, up-to-date, and clean data is a challenge.

- Data redundancy and gaps
- Unsupported formats
- Unclear permission and ownership issues of data



AI in Project Management: Managing Data

Data quality and availability

- Verify data sources
- Mindful of data limitations and biases
- Set-up alerts for anomalies
- Monitor and validate



AI in Project Management: Patterns of AI?

Seven (7) patterns of AI:

1. Conversational pattern
2. Recognition pattern
3. Anomaly detection pattern
4. Prediction analysis and decision support pattern
5. Hyper personalization pattern
6. Autonomous pattern
7. Goal driven pattern



AI in Project Management: AI Models

Choosing and Integrating AI Models

Picking the right AI model and integrating the existing workflows can be challenging.

- Different workflows
- Deployment challenges



AI in Project Management: AI Models

Choosing and Integrating AI Models

- Assess project needs
- Set clear expectations
- Pilot test
- Collaborate
- Road map



AI Tools in Project Management

High level Perspective



AI tools in Project Management

1. Tools Available for Project Management
2. AI implementation in projects



AI tools in Project Management

1. Machine Learning & Deep Learning
2. Natural Language Processing (NLP)
3. Predictive Analytics
4. Robotics Process Automation(RPA)



AI tools in Project Management

Impact of AI technologies for Project Managers

- Project Planning & Scheduling
 - Resource Management
 - Risk Management
 - Data Driven Decision-making
-
- Team collaboration and communication (reporting, meetings records, etc)
 - Task Automation
 - Predictive Analytics & Forecasting



AI tools in Project Management

Skills needed for Project Managers for AI technologies

- Knowledge
- AI literacy
- Ethical awareness
- Tools and Technologies available
- Capabilities
- Cost & Benefit Analysis
- Communication and Collaborating



AI in Project Management: GenAI

Generative AI offers tools to support Project Managers that can:

- Enhance productivity
- Improve project delivery success
- Help realize benefits more
- Open up new opportunities



AI in Project Management: GenAI benefits

- Save time
- Make informed decisions
- Adapt easily
- Offer versatility
- Gain competitive edge
- Boost efficiency
- Provide holistic view



AI Tools in Project Management

AI tools

GenAI Tools
(LLMs)

AI Agent
workflows

Client specific
tools
(Customization)



PM knowledge areas and suitable types of AI tools.

Item	PM knowledge area	Dominant knowledge type	Most suitable types of AI tools	Examples of AI capabilities
1	Integration management	- Formal - Data Driven - Tacit	GenAI AI Agents (LLMs & Integration)	Decision-making support based on real-time tracking and deviation alerts.
2	Scope management	- Data Driven - Tacit	GenAI AI Agents (LLMs & Integration)	Drafting work breakdown structures, clarifying deliverables
3	Schedule management	- Formal - Data Driven	ML for optimization AI Agents (LLMs & Integration)	Timeline generation, scheduling, forecasting, Critical Path identification.
4	Cost management	- Formal - Data Driven	ML for optimization AI Agents (LLMs & Integration)	Budget forecasting, variance analysis, EVM
5	Quality management	- Formal - Data Driven	ML for optimization AI Agents (LLMs & Integration)	Quality monitoring, anomaly detection.
6	Resource management	- Formal - Data Driven	ML for optimization AI Agents (LLMs & Integration)	Resource allocation, workload balancing.
7	Communication management	- Formal - Tacit	GenAI AI Agents (LLMs & Integration)	Summarizing meetings, drafting reports, language translation.
8	Risk management	- Data Driven - Tacit	GenAI ML for optimization AI Agents (LLMs & Integration)	Identifying emerging risks, probabilistic impact analysis.
9	Procurement management	Formal	Automation AI Agents (LLMs & Integration)	Contract analysis, vendor, sub-con selections.
10	Stakeholder management	- Formal - Tacit	GenAI AI Agents (LLMs & Integration)	Drafting engagement plans, analyzing feedback, conflict resolution assistance.



Student Exercise 1

- Perform a research to find and list the available AI tools which may be used in the project management processes.
 - Collating the business documentation.
 - Reviewing the lessons from the past projects.
 - Data quality check and filtering.
 - Planning and scheduling activities.
 - Resource Management.
 - Risk Management.
 - Cost and EVM (Earned Value Management).
 - Project Reporting.
 - Project Communication.
 - Any other tool which you may feel could be used.

Prepare your list and submit via
Submission Link Provided Separately.



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End of class



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Batam, Indonesia

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